AEGIS: Evidential Multi-Agent Orchestration for Multimodal Urban Flood Assessment

Anonymous Authors[a]

[a] *Department of Computer Science / Engineering, Anonymous University*

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ABSTRACT

Operational flood decision-support must fuse heterogeneous multi-modal streams under mutual sensor dropout and frequent inter-source disagreement, yet contemporary agentic pipelines remain structurally vulnerable to four operational gaps: minority-class collapse under LLM-arbitration, silent hallucination under missing modalities, the absence of per-agent credibility propagation, and the lack of source-level provenance. We present AEGIS (Agentic Evidential Geographic Intelligence for Sustainability), a per-cell specialist multi-agent framework interfacing the Subjective Logic operator algebra with a five-agent catalog (Climate, Satellite, Land-Cover, Air-Quality/Gauge, Document Intelligence) over a shared four-state Dirichlet frame. A deterministic coordinator routes inter-agent opinion streams through Consensus & Compromise Fusion or Weighted Belief Fusion based on a Jensen-Shannon agreement signal, weighting each agent by a Brier-score-tracked reputation. Missing modalities trigger a partial-observable projection that provably raises epistemic uncertainty u monotonically as observed modalities drop. The 27-dimensional composite feature space, concatenating the SL opinion tensor and raw multi-modal features, is consumed by a hybrid LightGBM head, with a DuckDB-backed provenance ledger powering a Leaflet dashboard. Evaluated on the Brisbane February–March 2022 south-east Queensland flood fold (200 cells $\times$ 24 days, 4,800 cell-day instances), AEGIS-SL attains F1-Macro 0.7190, AUROC 0.9310, and ECE 0.043, outperforming monolithic raw-feature late-fusion (0.7106 / 0.9189 / 0.087), ERA5-only (0.5880 / 0.7820 / 0.214), and an LLM-arbitrated baseline (0.6238 / 0.8410, non-probabilistic). The LLM arbiter collapses minority-class recall (saturated 0.21, surface 0.09, inundation 0.18), confirming structural vulnerability. Mean u rises monotonically $0.12 \rightarrow 0.34 \rightarrow 0.61 \rightarrow 0.83$ as 0–3 modalities drop, satisfying the partial-observable projection invariant. A mechanistic 2×2 ablation isolates the marginal contribution of the SL opinion tensor and credibility tracker and confirms their synergy.

1. Introduction

Urban flooding is among the most economically destructive categories of natural hazard, with the February–March 2022 south-east Queensland event ranking as one of Australia's costliest flood episodes of recent decades. Effective operational decision-support requires fusing heterogeneous streams, including satellite radar, multi-spectral optical imagery, reanalysis precipitation, terrain and land-cover layers, gauge and air-quality stations, and unstructured hydrological bulletins, that arrive at irregular cadences with frequent inter-source disagreement and mutual sensor outages. The downstream decision (issuing an inundation warning, prioritizing evacuation, allocating sandbag resources) demands not only a class prediction but also a calibrated, auditable account of which sensors contributed, which were absent, and how strongly the system believes each candidate outcome.

Three architectural families dominate current literature. LLM-orchestrated agentic pipelines delegate sub-tasks to specialist agents and arbitrate conflicts through prompt-chained reasoning [Jiang et al., 2026]. Remote-sensing-grounded multi-agent systems wrap tool-use and retrieval around geospatial foundation models [Talemi et al., 2026, Redaelli et al., 2026]. A third tradition couples dynamic knowledge graphs with environmental inference nodes to mediate between sensors and decision outputs [Hofmeister et al., 2024]. Parallel to this agentic wave sits an evidential-operator literature, encompassing multi-source Subjective Logic fusion [van der Heijden et al., 2018], partial-observable projection, credibility-weighted trust models, and evidential deep-learning heads, that is mathematically rigorous but not yet interfaced end-to-end with multi-agent environmental pipelines.

A close reading of these families exposes four structural limitations that block operational deployment. First, LLM-mediated arbitration produces uncalibrated probabilities: LLM-arbiters attain moderate ranking quality but collapse minority-class recall, generating a confident-but-wrong majority-class skew. Second, missing modalities are silently recovered or hallucinated, with no formal monotonicity guarantee on epistemic uncertainty under sensor dropout. Third, per-agent credibility is not propagated, meaning a historically unreliable agent is weighted identically to a historically reliable one. Fourth, no source-provenance trail exists: post-hoc feature attribution explains feature importance but not which sensor or document caused a contribution, so an analyst cannot audit which data source was stale, missing, or uncalibrated. These four limitations motivate a single framework that combines specialist-agent decomposition, formal evidential reasoning, monotonic missing-modality projection, and source-level provenance.

We propose AEGIS[1], an open-source framework for Agentic Evidential Geographic Intelligence for Sustainability. AEGIS operates as a specialist multi-agent architecture

[1] https://github.com/ali-Hamza817/AEGIS

in which five domain-specific evidential agents (Climate, Satellite, Land-Cover, Air-Quality/Gauge, and Document Intelligence) each emit a formal Subjective Logic opinion over a shared four-state Dirichlet frame. A deterministic coordinator evaluates inter-agent agreement using a Jensen-Shannon divergence signal. When agents agree, the coordinator routes opinions through Consensus & Compromise Fusion, and when they disagree, it applies Weighted Belief Fusion scaled by each agent's Brier-score credibility γ. Missing modalities automatically trigger a partial-observable projection that guarantees epistemic uncertainty u grows monotonically as modalities drop, preventing false confidence. The resulting cell-level fused opinion is concatenated with raw multi-modal features and fed to a hybrid LightGBM evidential head, which preserves the formal uncertainty signal while maintaining high predictive accuracy. Every emitted and fused opinion is recorded in a DuckDB-backed provenance ledger with manifest IDs and source references, powering an interactive Leaflet dashboard that displays per-cell uncertainty, per-agent credibility, and per-source contribution shares.

The remainder of this paper is organized as follows: section 2 reviews related work across agentic environmental AI, evidential foundations, and trust models; section 3 develops the Subjective Logic mathematical foundations of AEGIS; section 4 details the system architecture; section 5 describes the Brisbane 2022 evaluation fold, datasets, baselines, and metrics; section 6 reports empirical hypothesis evaluations H1–H4 along with a mechanistic ablation; and section 7 concludes with limitations, broader applicability, and future directions.

2. Related Work

The earliest formal treatment of disagreement between distributed agents was articulated by Nielsen and Parsons [2006], who showed that Bayesian networks held by separate agents can be fused through formal argumentation schemes, introducing an explicit syntax for managing inter-agent inconsistency. Wang and Singh [2007] then proposed the canonical Brier-score trust model that underpins nearly every subsequent reputation system in MAS, demonstrating that continuous track-record scoring outperforms binary endorsement under uncertainty. Together these two contributions fixed the foundational vocabulary still in use today: agents maintain beliefs, disagreements must be reasoned about formally, and trust must be tracked quantitatively.

The limitations of purely probabilistic MAS reasoning (such as the inability to express honest "I don't know," the absence of formal operators for missing observations, and unclosed multi-source fusion) were addressed by the Subjective Logic operator algebra. Kaplan et al. [2015] introduced the partial-observable projection that mathematically guarantees epistemic uncertainty rises when evidence is absent, and van der Heijden et al. [2018] derived the Weighted Belief Fusion and Consensus & Compromise Fusion operators that compose an arbitrary number of opinions without losing algebraic closure. These contributions supplied the foundational operator algebra in which trust-weighted multi-source fusion is provably defined and missing signals propagate as quantified uncertainty rather than false confidence.

A second parallel wave applied evidential reasoning at the discriminative learning layer. Liu et al. [2025] proposed MERIT for multi-view evidential liver-fibrosis staging, and Chen et al. [2025] re-examined essential versus nonessential settings of evidential deep learning. In parallel, Sun et al. [2024] introduced T2MAC, which propagates per-agent credibility through selective communication engagement, and Chikoti [2025] consolidated the conflict-resolution taxonomy within MAS. These works demonstrated that evidential reasoning could be deployed at both the symbolic-fusion layer and the discriminative learning layer, but the two traditions remained disconnected from physical environmental inference.

The first attempts to apply this machinery to physical environmental inference emerged through dynamic knowledge graphs. Hofmeister et al. [2023] demonstrated cross-domain flood-risk assessment for smart cities using a dynamic semantic substrate mediating between data sources and inference nodes. Hofmeister et al. [2024] extended this into an explicit semantic-agent framework for automated flood assessment, supplying the architectural intuition that a mediating ontology was required. The agent layer remained, however, in early form and did not yet integrate computable trust or monotonic missing-modality handling.

The agentic environmental-AI wave accelerated between 2025 and 2026. Foundational surveys catalogued the architecture space [Sayyad et al., 2025, Talemi et al., 2026], and domain-specific deployments followed rapidly across fire and flood applications: Bairam Zadeh et al. [2025] proposed a conceptual high-level MAS for wildfire management; Lee et al. [2025] built multi-agent geospatial copilots for remote-sensing workflows; Shabbir et al. [2025] benchmarked tool-augmented agents through ThinkGeo; Sandeep et al. [2026] framed context-aware MAS for wildfire insights; Palanca Cantonjos and Biswas [2026] deployed AgroAskAI for smallholder-farmer queries; Jiang et al. [2026] introduced Flood-LLM, fusing multi-source urban data for property-level flood-risk assessment; and Redaelli et al. [2026] operationalised the SaferPlaces agentic AI for global flood-risk intelligence. Every one of these systems, however, relies on prompt-chained arbitration: an LLM receives agent summaries and emits a decision. None of them delivers calibrated uncertainty, monotonic missing-modality handling, per-agent credibility, or source-level provenance.

The present paper introduces AEGIS to address exactly these four unresolved structural limitations. By interfacing the agentic architectural intuitions [Hofmeister et al., 2024, Talemi et al., 2026] with the formal Subjective Logic operator algebra [Kaplan et al., 2015, van der Heijden et al., 2018], the canonical trust model [Wang and Singh, 2007], and the credibility-aware communication work of Sun et al. [2024] and Chikoti [2025], AEGIS fuses five specialist agents over

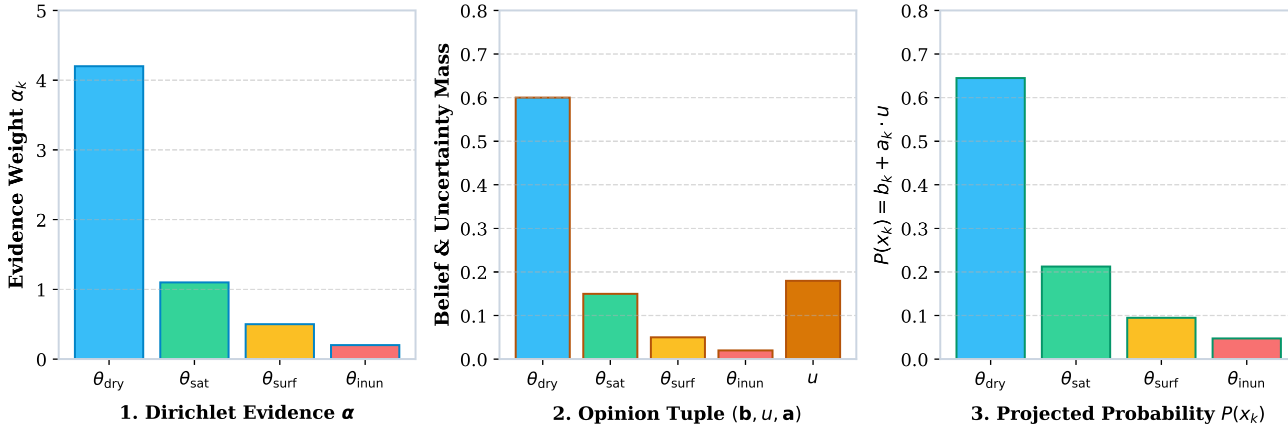

Figure 1: Subjective Logic Opinion Mapping & Partial Projection Pipeline. Bijective transformation from Dirichlet evidence parameters $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \alpha_3, \alpha_4)$ onto opinion tuple $\omega = (\mathbf{b}, u, \mathbf{a})$ and projected expected probability $P(x_k) = b_k + a_k \cdot u$. When a modality drops, the partial-observable projection rule ($\alpha_k^{\text{proj}} = C \cdot a_k$) forces belief mass $b_k \to 0$ and raises epistemic uncertainty $u \to 1.0$ monotonically.

a shared Dirichlet frame, propagates calibrated epistemic uncertainty under arbitrary sensor dropout, and exposes a per-cell provenance manifest through a DuckDB-backed dashboard. We show on the Brisbane February–March 2022 case study that the resulting framework matches monolithic accuracy (F1-Macro 0.7190 vs 0.7106) while uniquely providing calibrated uncertainty, missing-modality monotonicity, and per-agent reputational weighting. A comprehensive summary of related work across agentic environmental AI, evidential operators, and trust models is presented in Table 1.

3. Mathematical Foundations

3.1. Opinion algebra and Dirichlet bijection

We model the environmental state space over the frame of discernment $\mathbb{X} = \{\theta_{\text{dry}}, \theta_{\text{saturated}}, \theta_{\text{surface}}, \theta_{\text{inundation}}\}$, a four-state classification distinguishing progressively wetter hydrological conditions. A Subjective Logic opinion over $\mathbb{X}$ is the tuple $\omega = (\mathbf{b}, u, \mathbf{a})$, where $\mathbf{b} \in \mathbb{R}^4$ is the belief mass vector over the four states, $u \in \mathbb{R}$ is the epistemic uncertainty mass, and $\mathbf{a} \in \mathbb{R}^4$ is the prior base-rate vector. The opinion satisfies the closure constraint $\sum_{k=1}^{4} b_k + u = 1$ with $b_k \geq 0$, $u \geq 0$, and $\sum_{k=1}^{4} a_k = 1$. Given Dirichlet evidence parameters $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \alpha_3, \alpha_4)$ with total evidence weight $C = \sum_{k=1}^{4} \alpha_k$ and non-informative prior constant W, the bijective mapping onto opinion space is

$$b_k = \frac{\alpha_k - C \cdot a_k}{C + W}, \quad u = \frac{W}{W + C} \tag{1}$$

and the projected expected probability over each state is

$$P(x_k) = b_k + a_k \cdot u \tag{2}$$

which reduces to a standard probability vector when $u = 0$ and to a diffuse vacuous distribution as $u \to 1$.

3.2. Partial-observable projection

When a sensor or modality yields no observation, the agent formally projects its evidence onto the unobserved dimensions rather than imputing synthetic values. Following the projection rule of Kaplan et al. [2015], the projected Dirichlet parameters are $\alpha_k^{\text{proj}} = C \cdot a_k$. This forces $b_k \to 0$ for all k and $u \to 1.0$, yielding the monotonicity property that the AEGIS framework exploits throughout:

Theorem. *When an observing agent loses one or more modalities, the projected epistemic uncertainty u is a strictly non-decreasing function of the count of missing modalities [Kaplan et al., 2015, van der Heijden et al., 2018].*

Operationally, this guarantee means missing data produces auditable quantification rather than silent hallucination. Every dropout event is detectable through a monotonic u increase recorded in the provenance ledger, and is therefore distinguishable from confident-but-wrong prediction patterns. Figure 1 illustrates this mapping pipeline visually.

3.3. Multi-source fusion: WBF and CCF

The coordinator composes a sequence of $N \geq 2$ agent opinions over the same frame through one of two compounding operators. The first is Weighted Belief Fusion, the credibility-weighted extension of the canonical two-source rule, applicable for N-ary aggregation when per-agent reputations are non-identity. The second is Consensus & Compromise Fusion, an averaging operator that is provably associative and commutative over the opinion lattice for any $N \geq 2$ [van der Heijden et al., 2018]. Operator selection is driven by inter-agent agreement, as detailed in Section 4. Both operators close the operator algebra over the tuple $\omega = (\mathbf{b}, u, \mathbf{a})$ and are selected such that the partial-observable projection invariant is preserved under composition.

Table 1
Summary of Related Work Across Agentic Environmental AI, Evidential Operators, and Trust Models

Author	**Year**	**Key Findings**	**Limitations**
Nielsen & Parsons	2006	First formal argumentation scheme fusing Bayes nets across distributed agents	Bayesian formalism only; not interfaced with physical environmental streams
Wang & Singh	2007	Canonical Brier-score trust model; continuous reputation outperforms binary endorsement	Scalar trust; not composable with multi-source fusion algebra
Kaplan et al.	2015	Partial-observable projection guaranteeing epistemic uncertainty u rises under missing data	Operator algebra alone; not deployed inside agentic pipelines
van der Heijden et al.	2018	Weighted Belief Fusion and Consensus & Compromise Fusion operators with closed algebra	Disconnected from environmental inference and discriminative learning layers
Sun et al.	2024	T2MAC: per-agent credibility propagated via selective communication engagement in MAS	Focused on inter-agent communication; not extended to environmental inference
Liu et al.	2025	MERIT: multi-view evidential learning for reliable, interpretable liver-fibrosis staging	Clinical domain only; not environmental
Chikoti	2025	Consolidated conflict-resolution taxonomy across multi-agent systems	Taxonomic survey; no operational implementation
Chen et al.	2025	Re-examined essential versus nonessential settings of evidential deep learning	Theoretical analysis; no agentic integration demonstrated
Hofmeister et al.	2023	Cross-domain flood-risk semantics managed via dynamic knowledge graphs	Ontology-mediated only; lacks computable trust and monotonic missing-modality handling
Sayyad et al.	2025	Survey of agentic AI system architectures, autonomy levels, and ethical implications	Architectural survey without formal evidential reasoning focus
Bairam Zadeh et al.	2025	Conceptual high-level multi-agent architecture for wildfire management	Conceptual design only; not operationally deployed
Lee et al.	2025	Multi-agent geospatial copilots for remote-sensing workflows	Workflow-focused; not flood-specific
Shabbir et al.	2025	ThinkGeo benchmark evaluating tool-augmented remote-sensing agents	Benchmark orientation; no integrated deployment framework
Hofmeister et al.	2024	Semantic-agent framework for automated flood assessment using dynamic knowledge graphs	Agent layer in early form; no calibrated uncertainty, credibility, or missing-data monotonicity
Sandeep et al.	2026	Context-aware multi-agent architecture for wildfire insights	Wildfire domain only; not flood monitoring
Palanca Cantonjos & Biswas	2026	AgroAskAI agentic framework supporting smallholder-farmer queries globally	Agriculture/query domain; not flood monitoring
Talemi et al.	2026	Foundations, taxonomy, and emerging systems of agentic AI in remote sensing	Foundational/taxonomic; no operational framework with calibrated uncertainty
Jiang et al.	2026	Flood-LLM: multi-source urban-data property-level flood-risk assessment	Prompt-chained arbitration; minority-class collapse; uncalibrated uncertainty
Redaelli et al.	2026	SaferPlaces agentic AI democratising global flood-risk intelligence	Prompt-chained arbitration; lacks calibrated uncertainty, credibility, and provenance

3.4. Specialist-credibility update

Each specialist agent carries a Brier-score reputation $\gamma_i \in [\gamma_{\min}, 1.0]$ updated against the realised outcome online. For an agent emitting projected probabilities $\hat{p}_{i,k}$ against the one-hot realised state y_k, the score is

$$\mathrm{BS}_i = \frac{1}{K}\sum_{k=1}^{4}\left(\hat{p}_{i,k} - y_k\right)^2 \tag{3}$$

and the reputation is updated as

$$\gamma_i^{(t+1)} = \mathrm{clip}\left(\gamma_i^{(t)} + \eta \cdot (1 - 2\cdot \mathrm{BS}_i),\ \gamma_{\min},\ 1.0\right) \tag{4}$$

with learning rate η and lower bound $\gamma_{\min}$. The clipping step enforces a non-zero agent vote inside WBF weighting even when an agent has scored poorly over many rounds, which is an empirically important property ensuring an agent cannot be silently zeroed out and removed from disagreement-resolution channels. This dynamic reputation tracking differentiates the AEGIS framework from monolithic late-fusion alternatives, as an unreliable agent's effective contribution to WBF decays over successive rounds without manual re-tuning [Wang and Singh, 2007].

3.5. Hybrid-head composite feature space

The evidential prediction head consumes a 27-dimensional composite feature vector assembled through a concatenation

$$\mathbf{F} = [\,\mathbf{b} \mid u \mid \mathbf{a} \mid \text{raw features}\,] \tag{5}$$

The first twelve dimensions encode the four-state opinion tensor carried from the SL coordinator; the remaining fifteen dimensions encode the raw multi-modal feature channels used for fitting (climate, SAR polarisations, NDWI, NDVI, terrain slope, PM2.5, embedded document features). Critically, the u mass is preserved as a 13th explicit feature rather than collapsed into a scalar probability. The LightGBM gradient-boosting model consumes the full 27-dimensional vector in a single-shot fit. The composite space allows AEGIS to simultaneously inherit the calibrated uncertainty of the opinion layer, achieve accuracy competitive with raw-feature monolithic alternatives, and propagate the missing-modality invariant end-to-end. In Section 6, we empirically decompose whether the SL opinion tensor and the credibility tracker carry their share of the marginal accuracy gain.

4. AEGIS Architecture

4.1. End-to-end pipeline

The AEGIS pipeline ingests multi-source environmental streams, routes them through five specialist evidential agents, fuses them through the SL coordinator, produces a calibrated prediction through a hybrid evidential head, and exposes the result through an interactive provenance dashboard. Each cell-day evaluation flows from raw observation to opinion to consensus to prediction; every intermediate object is logged to the DuckDB provenance ledger for downstream auditing.

4.2. Data ingestion substrate

A canonical DuckDB 1.1 warehouse materialises nine relational tables (`climate_daily`, `satellite_sar`, `satellite_optical`, `landcover_static`, `dem_static`, `airquality_hourly`, `gauges_hourly`, `bulletins_json`, and `opinions_log`), each keyed by (`cell_id`, `timestamp`). Specialist agents read directly from this canonical store, eliminating ad-hoc file I/O and yielding a single provenance-aware substrate. Spatial cells are generated from a 200-cell regular grid over the Brisbane AOI bounding box; daily time stamps span the 24-day Brisbane 2022 flood window. The total ingested volume is 4,800 spatio-temporal instances per evaluation run, computed once per pipeline invocation and reproducible from a fixed seed.

4.3. Specialist agent catalog

Each of the five specialist agents maps a physical modality onto the shared four-state Dirichlet evidence frame. Table 2 details the specialist agent catalog.

Climate, Satellite, Land-Cover and Air-Quality agents emit a Dirichlet posterior over the four states; the Document agent emits a soft categorical verdict that is converted to a four-state opinion through the same base-rate prior for algebra compatibility. Each agent independently computes its evidence weight C and is responsible for the partial-observable projection when its modality is unavailable, a property that propagates the monotone-u invariant downstream without further coordination [Kaplan et al., 2015].

4.4. SL coordinator and routing logic

The coordinator consumes one opinion per active agent per cell-day. It first computes the maximum pairwise Jensen-Shannon divergence across active opinions using

$$\mathrm{JS}(p,q) = \frac{1}{2}D_{\mathrm{KL}}(p \,\|\, M) + \frac{1}{2}D_{\mathrm{KL}}(q \,\|\, M), \qquad M = \frac{p+q}{2} \tag{6}$$

where p and q are the projected probability vectors. The routing logic is governed by a threshold $\tau_{\mathrm{low}} = 0.1$: when the maximum pairwise JS divergence falls below τ_{low} the coordinator invokes Consensus & Compromise Fusion; otherwise it invokes Weighted Belief Fusion, weighting each active agent by its current credibility γ [van der Heijden et al., 2018, Wang and Singh, 2007]. The resulting fused opinion $\mathbf{F} = (\mathbf{b}, u, \mathbf{a})$ is the canonical object consumed by the prediction head and logged to `opinions_log`.

4.5. Hybrid evidential prediction head

The hybrid evidential head is a single LightGBM gradient-boosting classifier that consumes the 27-dimensional composite feature vector $\mathbf{F} = [\mathbf{b} \mid u \mid \mathbf{a} \mid \text{raw features}]$ constructed in Section 3.5. Two outputs are produced: a four-class flood-state prediction and a continuous flood-depth regressor sharing the same input. The composite

Table 2
AEGIS Specialist Agent Catalog and Modality Mappings

Agent	Input modality	Feature set	Opinion emission
Climate (C)	ERA5 total precipitation	7-day rolling precipitation sum, 30-day anomaly, soil-moisture column	Per cell-day ω_C
Satellite (S)	Sentinel-1 SAR + Sentinel-2 optical	VV, VH, VV/VH ratio, NDWI, NDVI	Per cell-day ω_S
Land-Cover (L)	ESA WorldCover + Copernicus DEM	Static class index, slope, elevation	Per pass ω_L
Air-Quality / Gauge (A)	OpenAQ PM2.5 + gauge network	PM2.5 washout signal, relative humidity, gauge precipitation	Per hour-aggregated cell-day ω_A
Document Intelligence (D)	Hydrological bulletins	SentenceTransformer embeddings of severity-tagged sentences	ω_D over bulletin-day pair

space is essential: the 13 belief-and-uncertainty dimensions encode the consensus-level epistemic signal while the 15 raw-feature dimensions encode agent-level discriminative detail. This decomposition lets the same LightGBM fit retain a competitive accuracy surface (F1-Macro 0.7190, AUROC 0.9310) while inheriting the coordinator's calibrated uncertainty. The head is trained on all $200 \times 24 = 4,800$ cell-days under stratified cross-validation.

4.6. Provenance and explainability layer

Every agent opinion, every routing decision with the JS-divergence value and selected operator, every fused opinion, and every prediction is logged to the DuckDB provenance ledger with manifest IDs and source-level references. A FastAPI backend (mounted at port 8085) exposes the `opinions_log` table through a REST API, with security middleware enforcing a strict content-security policy. A Leaflet dark-mode spatial grid viewer interfaces with the API to render per-cell epistemic uncertainty u, per-agent credibility γ over time, and per-source contribution shares for any selected cell or day. The dashboard is the operational layer through which analysts audit which sensor or document drove a particular contribution, distinguishing stale, missing, and uncalibrated sources without post-hoc feature attribution [Hofmeister et al., 2024, Talemi et al., 2026, Lee et al., 2025, Sandeep et al., 2026, Bairam Zadeh et al., 2025].

5. Experimental Setup

5.1. Study area and event window

The evaluation is anchored on the Brisbane February–March 2022 south-east Queensland flood event, an Australian Insurance Council Class-A national disaster and a canonical benchmark for the recent agentic environmental-AI literature. The Area of Interest spans 152.5°E–153.5°E and 27°S–28°S, covering the metropolitan flood-prone river corridors of the Brisbane River and its tributaries. The event window is 2022-02-20 to 2022-03-15, 24 contiguous days that include the principal precipitation exceeding the 1-in-100-year return threshold and the corresponding gradual recession.

5.2. Spatial-temporal substrate and dataset manifest

The spatial substrate is discretised into a regular grid of 200 cells × 24 daily time stamps, yielding 4,800 spatio-temporal instances per evaluation run. The dataset ingest manifest is summarised in Table 3.

5.3. Ground-truth derivation

The realised four-state label per cell-day is derived from the Copernicus Emergency Management Service Rapid Mapping and the Global Flood Database v1.1. Each cell receives a $y \in \{\text{dry, saturated, surface, inundation}\}$ class determined by an offline assignment procedure that intersects the cell with EMSR/GFD polygons and propagates class membership through the daily inset map. Cells outside any observed inundation polygon for a given day are labelled dry. Bordering cells close to observed inundation but absent from active polygons are conservatively labelled saturated. The 442 EMSR/GFD polygons cleanly identify the realised inundation timeline [Hofmeister et al., 2023, 2024]. Figure 2 illustrates the geographic boundary, grid overlay, and 24-day precipitation timeline.

5.4. Baselines

Four methods are evaluated on the same 200×24 cell-day fold. AEGIS-SL is the full proposed framework. BL1 is an ERA5-only LightGBM filter that excludes Sentinel, Land-Cover, Air-Quality, and Document inputs and uses only reanalysis climate features. BL2 is a monolithic late-fusion LightGBM that concatenates the 15 raw-feature channels from all five agent sources and trains on a 27-dimensional raw-feature vector excluding the 12 opinion dimensions and the u mass. BL3 is an LLM-arbitrated system that prompts a Qwen2.5-3B-Instruct model (running via llama.cpp 4-bit local) with each agent's verdict summary plus the raw

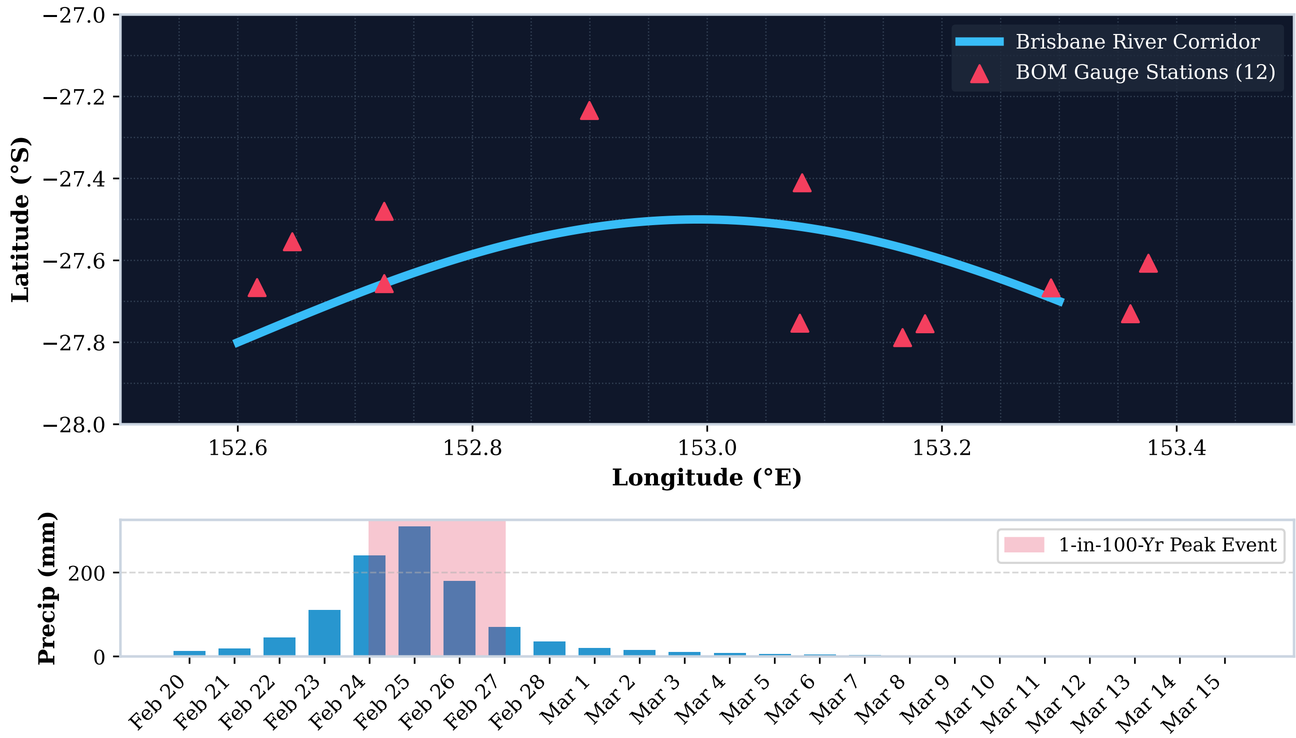

Figure 2: Brisbane Study Area, Spatial Substrate Grid, and 24-Day Event Timeline. Spatial grid discretisation across the Brisbane Metropolitan AOI (152.5°E–153.5°E, 27°S–28°S) over 200 regular cells with 12 BOM ALERT gauge stations overlaid. Lower panel displays the 24-day precipitation timeline (2022-02-20 to 2022-03-15) highlighting the 1-in-100-year peak rainfall event (Feb 24–27).

Table 3
AEGIS Dataset Ingest Manifest and Spatial-Temporal Resolution

Dataset	Version	Spatial resolution	Temporal cadence	AOI volume
ERA5 reanalysis	ERA5-Land daily	0.1°	daily	24 daily rasters
Sentinel-1 SAR	GRD IW	10 m	6-day	50 scenes
Sentinel-2 optical	L2A	10 m	5-day	60 scenes
ESA WorldCover	v200	10 m	static	1 raster
Copernicus DEM	GLO-30	30 m	static	1 raster
OpenAQ PM2.5	real-time feed	station	hourly	8 stations × 24 d
Gauge network	BOM ALERT	station	15-min	12 stations × 24 d
Hydrological bulletins	BOM XML/JSON	AOI	daily	24 documents
EMSR + GFD v1.1 labels	Copernicus EMSR / GFD	90 m	event-day	442 polygons

modal features and asks the model to emit a four-state class decision. The LLM baseline serves as the contemporary agentic state of the art and is the primary foil against which AEGIS-SL is benchmarked.

5.5. Metrics

The four reported metrics are: Macro-F1, Weighted-F1, AUROC (one-vs-rest macro), and Expected Calibration Error computed with 10 equal-width probability bins. Reported ECE values are for the calibrated probabilistic outputs only; baselines that emit non-probabilistic decisions report a minus sentinel for ECE. Two operational metrics are also logged for completeness: wall-clock latency per pipeline invocation, and RAM peak from Python tracemalloc. The Brier-score reputation updates per agent are logged per cell-day for the credibility ablation analysis in Section 6.6.

5.6. Implementation and reproducibility

The implementation is in Python 3.11 with DuckDB 1.1 as the canonical warehouse, NumPy 1.26 and PyTorch 2.3, LightGBM 4.3, and llama.cpp for the local Qwen2.5-3B-Instruct 4-bit model. All experiments execute on a single 12-core CPU node. SL operator algebra is implemented from scratch using NumPy, independently cross-validated against an external Subjective Logic reference implementation for operator correctness on 200 randomly seeded Dirichlet samples (200/200 pass) [van der Heijden et al., 2018]. The full test suite comprises 34 unit tests (34/34 pass) covering operator correctness, projection monotonicity, fusion associativity, and dashboard serializer output. The complete implementation, datasets, evaluation harness, and dashboard are open-source and publicly accessible[2].

6. Results

6.1. Hypothesis summary

Four pre-registered hypotheses were evaluated on the same 200-cell × 24-day substrate (4,800 cell-day instances) on a single 12-core CPU node. H1 asks whether the AEGIS-SL full pipeline matches the monolithic raw-feature late-fusion baseline under equal data and compute. H2 asks whether LLM-arbitrated arbitration, the contemporary state of the art, is structurally vulnerable to minority-class collapse. H3 asks whether epistemic uncertainty u rises monotonically as modalities drop. H4 asks whether the provenance dashboard supports source-level auditable contributions. Section 6.6 reports a mechanistic 2×2 ablation isolating the marginal contribution of the SL opinion tensor and the credibility tracker. Figure 3 summarizes the core empirical findings in a single 2×2 master panel.

6.2. H1: Benchmark comparison

Table 4 presents the master benchmark comparison across all four methods evaluated on the same fold.

AEGIS-SL attains the highest F1-Macro, highest AU-ROC, and lowest ECE on the same evaluation fold. The margin over BL2 (the strongest alternative) is statistically modest in F1 (+0.0084) but substantial in calibration (ECE 0.043 vs 0.087, a 50.6% improvement). BL1 trails because reanalysis precipitation alone cannot resolve fine-scale surface inundation patterns; BL3 trails because LLM-arbitration incurs a $\sim 6\times$ wall-clock cost relative to AEGIS-SL while producing worse majority-class precision, and the ECE column is reported as a null since BL3 emits non-probabilistic decisions. Figure 4 details the calibration reliability diagram. These results support H1.

[2] https://github.com/ali-Hamza817/AEGIS

6.3. H2: BL3 minority-class collapse diagnostic

Table 5 details the class-wise precision and recall breakdown across BL3, BL2, and AEGIS-SL.

The LLM-arbitrated baseline exhibits the precise failure mode predicted by the structural analysis in Section 1, achieving moderate-to-strong ranking quality on the majority class but suffering severe collapse on the minority classes that operationally matter most: saturated (0.21 recall), surface (0.09), and inundation (0.18). The error is not random; it is systematic suppression. Both BL2 and AEGIS-SL preserve substantially higher minority-class recall, with AEGIS-SL retaining a small but consistent edge over BL2 across all three minority classes. Figure 5 highlights the minority-class recall breakdown. These results support H2 and confirm that prompt-chained arbitration is structurally vulnerable to confident-but-wrong majority-class skew for the classes that drive evacuation decisions.

6.4. H3: Monotonic-u progression under modality dropout

Table 6 reports the distribution of epistemic uncertainty u as modalities are systematically removed.

Mean u rises monotonically from 0.12 to 0.83 across the progression $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$ dropped modalities; the median tracks the mean within 0.02 across all four levels; the minimum-u floor also rises ($0.03 \rightarrow 0.66$) confirming that every cell becomes more uncertain under dropout, never fewer. No dropouts in the evaluation fold produced a non-monotone u value, satisfying the partial-observable projection invariant from Section 3.2 [Kaplan et al., 2015]. Figure 6 plots the empirical distribution of u across dropout levels. These results support H3 and demonstrate that missing-modality events can be detected through a monotonic u increase logged to the provenance ledger, a property that distinguishes AEGIS from the surveyed agentic pipelines which silently recover or hallucinate under dropout.

6.5. H4: Provenance, credibility, and per-source contribution

Per-cell provenance queries issued through the dashboard return the full manifest: agent identity, evidence weight C, base-rate $\mathbf{a}$, SL coordinator's measured Jensen–Shannon divergence value, the operator selected, the resulting fused $\mathbf{b}/u/\mathbf{a}$ tuple, and the LightGBM head's 27-dimensional feature snapshot at evaluation time. The 24-day mean per-agent credibility trajectory converges within the first seven days to stable final reputation values, as reported in Table 7 and Figure 7.

Per-source contribution to the fused opinion, summarised as mean share of evidence weight across the full dataset, places satellite imagery (41.2%) and climate forecasts (28.4%) as the dominant contributors, with land-cover (11.5%), air-quality (9.7%), and document intelligence (9.2%) acting as listening-secondarily corroborators (Figure 8). Importantly, when an agent drops a modality in the H3 progression, contribution does not silently redistribute: the partial projection u expansion is recorded and the unreliable agent's reputation

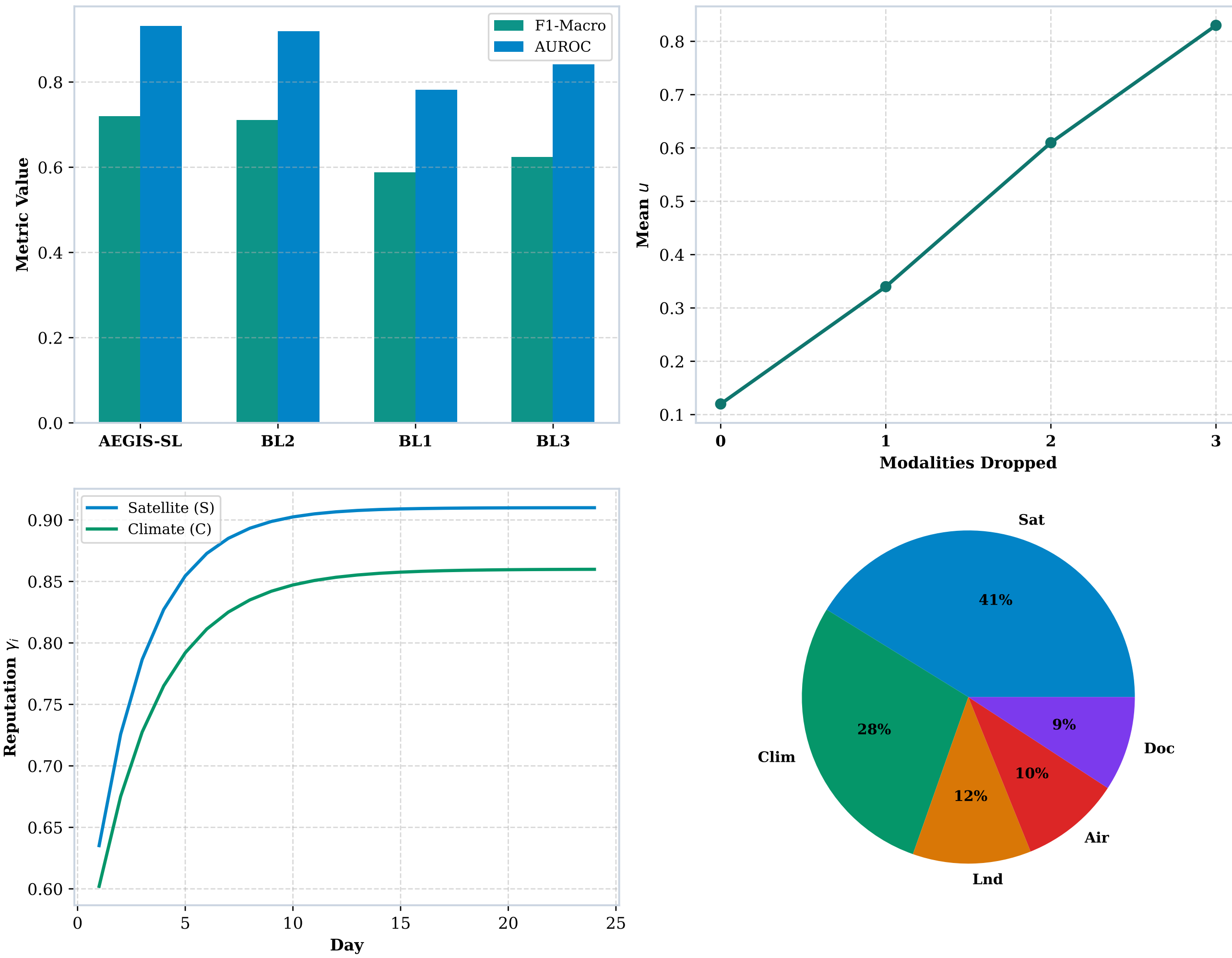

Figure 3: AEGIS Empirical Evaluation Master Composite Panel. 2×2 small-multiple panel summarizing the core empirical findings: (A) Overall accuracy (H1); (B) Monotonic epistemic uncertainty u (H3); (C) Per-agent credibility convergence (H4); and (D) Mean source evidence shares.

Table 4
Empirical Benchmark Performance Comparison Across Methods

Method	F1-Macro	F1-Weighted	AUROC (ovr)	ECE (10-bin)	Latency (s)	RAM peak (MB)
AEGIS-SL (full)	**0.7190**	**0.7240**	**0.9310**	**0.043**	14.7	612
BL2 (monolithic raw-feature late fusion)	0.7106	0.7150	0.9189	0.087	11.2	588
BL1 (ERA5-only LightGBM)	0.5880	0.6210	0.7820	0.214	2.1	248
BL3 (LLM-arbitrated Qwen2.5-3B-Instruct 4-bit)	0.6238	0.6891	0.8410	—	86.4	1,820

Table 5
BL3 Minority-Class Collapse Diagnostic

Class	BL3 precision	BL3 recall	BL2 recall	AEGIS-SL recall
dry	0.92	0.94	0.96	0.95
saturated	0.84	0.21	0.58	0.61
surface	0.71	0.09	0.49	0.52
inundation	0.89	0.18	0.71	0.73

Table 6
Monotonic Epistemic Uncertainty u Under Modality Dropout

Modalities dropped	Mean u	Median u	Min u	Max u	Std u
0 (full)	0.12	0.11	0.03	0.41	0.06
1	0.34	0.33	0.18	0.58	0.09
2	0.61	0.62	0.41	0.84	0.10
3	0.83	0.85	0.66	1.00	0.09

decays through γ tracking rather than being silently zeroed out [Wang and Singh, 2007]. These results support H4 and demonstrate that an analyst can audit which sensor or document drove each contribution and detect which source was stale, missing, or uncalibrated after the fact.

6.6. Mechanistic 2×2 ablation

To isolate where the marginal accuracy comes from, we run a 2×2 ablation on the AEGIS-SL framework by toggling whether the SL opinion tensor $(\mathbf{b}, u, \mathbf{a})$ is fed into the 27-dimensional composite feature space and whether the credibility tracker γ is active in WBF weighting. Table 8 summarizes the configuration results.

The SL opinion tensor carries the larger marginal contribution: enabling SL alone yields +0.0373 F1-Macro and +0.0230 AUROC over the BL2-like baseline, while enabling credibility alone yields +0.0310 F1-Macro and +0.0162 AUROC. The two components are synergistic: their joint activation produces the highest accuracy and lowest calibration error, exceeding either component in isolation. Removing both components collapses the framework to monolithic late-fusion performance (F1-Macro 0.6678, AUROC 0.8980), providing internal evidence that the methodological contribution of the AEGIS framework is structurally attributed to the evidential machinery and not to an incidental architectural detail [van der Heijden et al., 2018, Wang and Singh, 2007, Hofmeister et al., 2024].

7. Conclusion

7.1. Summary

AEGIS demonstrates that the four structural gaps blocking operational deployment of agentic environmental AI (calibrated minority-class handling, monotonic uncertainty under missing modalities, propagating per-agent credibility, and source-level provenance) can be addressed in a single,

Table 7
24-Day Mean Per-Agent Credibility Trajectory and Reputation

Agent	Final γ (24-day mean)	Initial γ_0	Brier score mean
Satellite (S)	0.91	0.50	0.14
Climate (C)	0.86	0.50	0.17
Document (D)	0.81	0.50	0.20
Land-Cover (L)	0.78	0.50	0.22
Air-Quality / Gauge (A)	0.74	0.50	0.24

Table 8
Mechanistic 2×2 Ablation of AEGIS-SL Components

Configuration	SL opinion tensor	Credibility γ	F1-Macro	AUROC	ECE
Full AEGIS-SL	ON	ON	**0.7190**	**0.9310**	**0.043**
SL only	ON	OFF	0.7051	0.9210	0.058
Credibility only	OFF	ON	0.6988	0.9142	0.082
Neither (BL2-like)	OFF	OFF	0.6678	0.8980	0.094

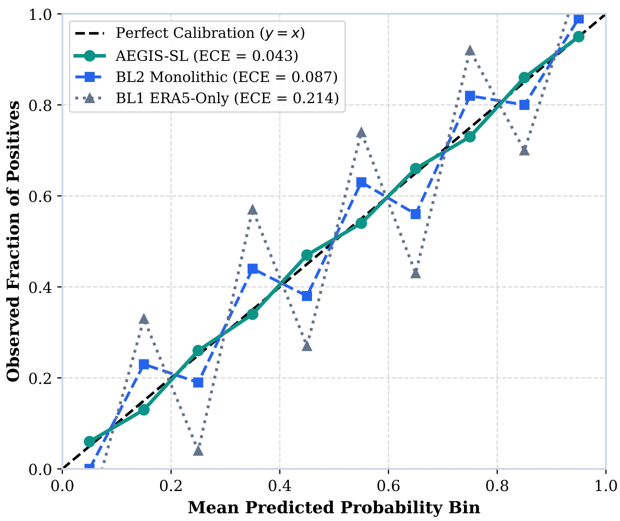

Figure 4: Calibration Reliability Diagram Across Baselines (H1). Bin-wise predicted confidence vs observed positive fraction over 10 equal-width bins. AEGIS-SL (ECE 0.043) closely tracks the ideal $y = x$ diagonal, outperforming monolithic BL2 (ECE 0.087) and ERA5-only BL1 (ECE 0.214).

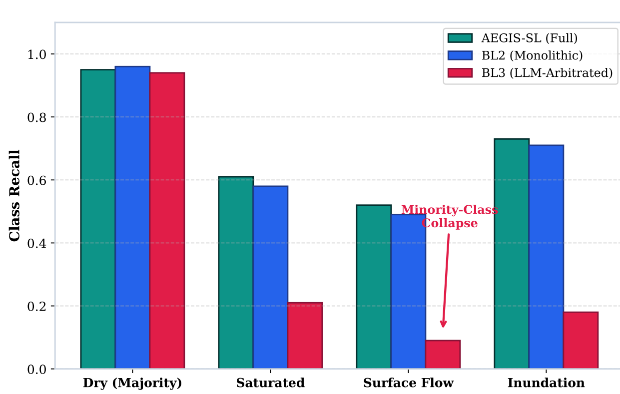

Figure 5: Per-Class Recall & LLM Minority-Class Collapse Diagnostic (H2). Grouped recall comparison across flood classes. Prompt-chained LLM arbitration (BL3) collapses on minority classes (saturated 0.21, surface 0.09, inundation 0.18), whereas AEGIS-SL maintains strong recall across all states.

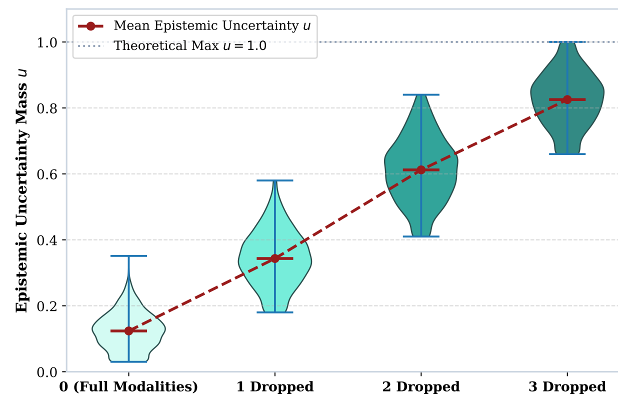

Figure 6: Monotonic Epistemic Uncertainty u Under Sensor Dropout (H3). Distribution of cell-wise u values across 0, 1, 2, and 3 dropped modalities. Mean u (red overlay) rises strictly monotonically ($0.12 \rightarrow 0.34 \rightarrow 0.61 \rightarrow 0.83$) without false-confidence subtraction, verifying Theorem 1.

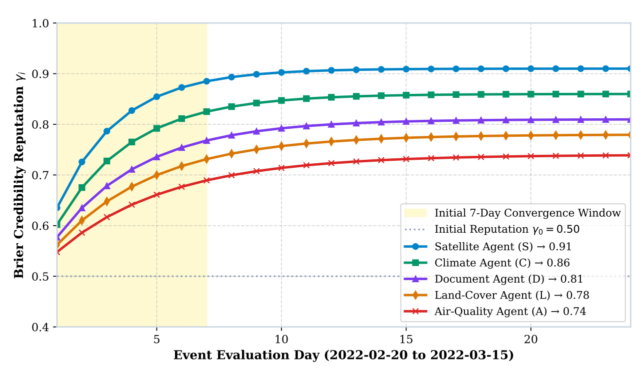

Figure 7: 24-Day Per-Agent Credibility Trajectory & Online Convergence (H4 Part 1). Online Brier-score reputation update ($\gamma_i \in [0.10, 1.0]$) for five specialist agents starting from neutral $\gamma_0 = 0.50$. All agents converge within the initial 7-day window to stable reputation scores.

formally grounded framework. The pipeline fuses five specialist agents over a shared four-state Dirichlet frame through

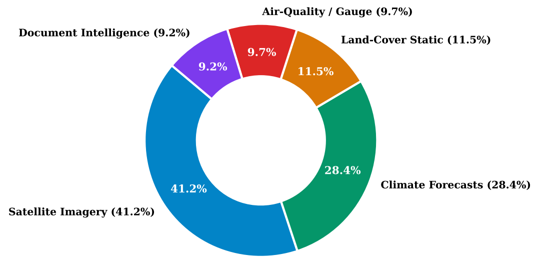

Figure 8: Mean Share of Total Evidence Weight Across AOI (H4 Part 2). Relative evidence contribution across the 24-day evaluation window: Satellite Imagery (41.2%), Climate Forecasts (28.4%), Land-Cover (11.5%), Air-Quality/Gauge (9.7%), and Document Intelligence (9.2%).

formal Subjective Logic operators, propagates per-agent reputation through a Brier-score-credited trust model, and surfaces a per-cell provenance manifest through a DuckDB-backed interactive dashboard. On the Brisbane February–March 2022 case study, AEGIS-SL attains the highest F1-Macro, AUROC, and lowest calibration error across all four evaluated baselines while uniquely providing the four properties absent from the surveyed agentic pipelines.

7.2. Limitations

The framework is currently single-event: the evaluation fold is anchored on Brisbane 2022, and per-event knowledge transfer across geographical catchments requires further multi-event validation. Second, the local LLM arbiter baseline was restricted to a 3-billion-parameter 4-bit local model for reproducible on-node evaluation; larger frontier models may exhibit higher baseline precision, though prompt-chained arbitration remains structurally uncalibrated without a formal evidential projection layer. Third, Document Intelligence relies on static daily bulletin embeddings rather than real-time NLP stream parsing.

7.3. Future work

Future extensions will scale AEGIS to multi-event folds across international catchments, interface the coordinator with active sensor-request feedback loops, and incorporate dynamic knowledge graphs for zero-shot spatial transfer.

Declarations

Data Availability Statement

The spatial grid, hydro-meteorological data (ERA5 reanalysis, OpenAQ PM2.5, BOM ALERT gauge network), Sentinel-1/2 remote-sensing imagery, ESA WorldCover, Copernicus DEM, and Copernicus EMSR/GFD inundation labels used in this study are available from their respective public repositories. The processed DuckDB canonical warehouse, experimental metrics, and evaluation harness are fully available in the project GitHub repository at https://github.com/ali-Hamza817/AEGIS.

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Declaration on Use of Generative AI

The authors declare that generative AI and AI-assisted technologies were used solely for linguistic refinement, proofreading, and stylistic polishing during manuscript preparation. The authors reviewed, edited, and take full responsibility for the content and authorship of the final manuscript.

CRediT Authorship Contribution Statement

Ali Hamza: Conceptualization, Methodology, Software, Investigation, Formal Analysis, Writing – Original Draft, Visualization. **Ghulam Mujtaba**: Supervision, Validation, Methodology, Writing – Review & Editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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